

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

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# Problem-Solving and Data Analysis

# 09

10 answers · Rationale-rich · Teach from doc

**Q1****B**

B. Theresa's speed was increasing for a longer period of time than it was decreasing.

Choice B is correct. Theresa's speed was increasing from 0 to 5 minutes and from 20 to 25 minutes, which is a total of 10 minutes. Theresa's speed was decreasing from 10 minutes to 20 minutes and from 25 to 30 minutes, which is a total of 15 minutes. Therefore, Theresa's speed was NOT increasing for a longer period of time than it was decreasing.

Choice A is incorrect. Theresa ran at a constant speed for the 5-minute period from 5 to 10 minutes. Choice C is incorrect. Theresa's speed decreased at a constant rate during the last 5 minutes, which can be seen since the graph is linear during that time. Choice D is incorrect. Theresa's speed reached its maximum at 25 minutes, which is within the last 10 minutes.

**Q2****31**

The correct answer is 31. It's given that 1 yard is equal to 36 inches. Therefore, 1,116 inches is equivalent to  $1,116 \text{ inches} \frac{1 \text{ yard}}{36 \text{ inches}}$ , or 31 yards.

**Q3****A**

A. 25%

Choice A is correct. Let  $x$  represent the percentage of 300 that is 75. This can be written as  $\frac{x}{100}300 = 75$ , or  $3x = 75$ . Dividing both sides of this equation by 3 yields  $x = 25$ . Therefore, 25% of 300 is 75.

Choice B is incorrect. 50% of 300 is 150, not 75.

Choice C is incorrect. 75% of 300 is 225, not 75.

Choice D is incorrect. 225% of 300 is 675, not 75.

**Q4****B****B.**  $\frac{9}{50}$ 

Choice B is correct. If one of these children is selected at random, the probability of selecting a child who chose a vegetarian sandwich is equal to the number of children who chose a vegetarian sandwich divided by the total number of children. According to the table, there are a total of 50 children, and 9 of these children chose a vegetarian sandwich. Therefore, the probability of selecting a child who chose a vegetarian sandwich is  $\frac{9}{50}$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q5****C****C.** 82

Choice C is correct. The table gives the results of 200 people who were asked how often they see a movie in a theater. The table shows that 29 people responded “never” and 53 people responded “almost never.” Therefore,  $29 + 53$ , or 82, people responded either “never” or “almost never.”

Choice A is incorrect. This is the difference between the number of people who responded “almost never” and the number of people who responded “never.”

Choice B is incorrect. This is the number of people who responded “almost never” but doesn't include those who responded “never.”

Choice D is incorrect. This is the number of people who responded something other than “never” or “almost never,” rather than the number of people who responded either “never” or “almost never.”

**Q6**

**B**

B. As the elevation increases, the temperature tends to decrease.

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Choice B is correct. The association between the elevation and the temperature of locations in the Lake Tahoe Basin can be described by looking at the direction of the line of best fit. The line of best fit slopes downward, which corresponds to the temperature decreasing as the elevation increases.

Choices A and C are incorrect. Both of these choices would be represented by a line of best fit that slopes from the lower left to the upper right of the graph, which isn't what's shown on the graph. Choice D is incorrect. This choice would be represented by a line of best fit that is horizontal or has a slope very close to 0. This is not what's shown on the graph.

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**Q7**

**9**

The correct answer is 9. It's given that the customer spent \$ 27 to purchase oranges at \$ 3 per pound. Therefore, the number of pounds of oranges the customer purchased is  $\$ 27 \frac{1 \text{ pound}}{\$ 3}$ , or 9 pounds.

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**Q8****C****C. 912**

Choice C is correct. According to the table, 63% of survey respondents get most of their medical information from a doctor and 13% get most of their medical information from the Internet. Therefore, 76% of the 1,200 survey respondents get their information from either a doctor or the Internet, and 76% of 1,200 is 912.

Choices A, B, and D are incorrect. According to the table, 76% of survey respondents get their information from either a doctor or the Internet. Choice A is incorrect because 865 is about 72% (the percent of survey respondents who get most of their medical information from a doctor or from magazines/brochures), not 76%, of 1,200. Choice B is incorrect because 887 is about 74%, not 76%, of 1,200. Choice D is incorrect because 926 is about 77%, not 76%, of 1,200.

**Q9****A****A. 0.004**

Choice A is correct. The probability of randomly selecting a defective MP3 player from the shipment is equal to the number of defective MP3 players divided by the total number of MP3 players in the shipment. Therefore, the probability is  $\frac{4}{1,000}$ , which is equivalent to 0.004.

Choice B is incorrect because 0.04 represents 4 defective MP3 players out of 100 rather than out of 1,000. Choice C is incorrect because 0.4 represents 4 defective MP3 players out of 10 rather than out of 1,000. Choice D is incorrect. This is the number of defective MP3 players in the shipment.

## Q10

C

C. 86

Choice C is correct. The mean of a data set is found by dividing the sum of the values in the data set by the number of values in the data set. Therefore, the mean of data set A is

$$\frac{72+73+73+76+76}{5}$$

, which simplifies to 74. The mean of data set B is represented by the

equation  $\frac{61+64+74+85+x}{5}$ , or  $\frac{284+x}{5}$ . It's given that the mean of data set A is equal to

the mean of data set B. Therefore, the equation  $74 = \frac{284+x}{5}$  can be used to solve for x.

Multiplying both sides of this equation by 5 yields  $370 = 284 + x$ . Subtracting 284 from both sides of this equation yields  $86 = x$ .

Choices A, B, and D are incorrect and may result from calculation errors.

### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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